

Physical Activity and Anxiety Levels Among High School Students at Doral Academy

Teacher: Kristina Zogović

Course: AP Statistics

School: Doral Academy Charter High School, Doral, Florida

Researchers: Isabel Aponte, Micaela Velez, Mia Bilinski

Date: February 25, 2026

Author Note

This research was completed in partial fulfillment of the AP Statistics course under the supervision of Prof. Kristina Zogović. Correspondence concerning this article should be addressed to the researchers via Doral Academy Charter High School, Doral, Florida.

Abstract

Among Doral Academy high school students, is there a statistically significant linear relationship between weekly physical activity (minutes per week) and anxiety level (GAD-7 score)? A stratified random sample of 60 students (15 per grade from 9–12) completed an anonymous survey. Least-squares regression yielded $\hat{y} = 21.973 - 0.04249x$, $r = -.958$, and $R^2 = .918$. The slope test produced $t(59) = -25.43$, $p < .0001$. These results provide strong evidence that higher weekly physical activity is associated with lower anxiety scores.

Keywords: physical activity, anxiety, GAD-7, adolescents, high school students

Introduction

Adolescent anxiety has become a pressing public health issue, influencing sleep quality, academic performance, concentration, and overall wellbeing. Physical activity is frequently

recommended as a beneficial, accessible strategy to support mental health. The Centers for Disease Control and Prevention (2024) recommends that adolescents engage in at least 60 minutes of moderate-to-vigorous physical activity daily, yet compliance remains low in many student populations.

This study focused on students at Doral Academy Charter High School, where rigorous academics, extracurricular activities, and screen time may elevate stress levels.

Research Question

Among Doral Academy high school students, is there a statistically significant linear relationship between weekly physical activity (minutes per week) and anxiety level as measured by the GAD-7 score?

Method

Participants

Participants were 60 high school students (15 from each grade 9–12) selected using stratified random sampling from Doral Academy Charter High School. This method ensured balanced representation across grade levels.

Procedure

Participation was voluntary and anonymous. Students were informed of their right to withdraw at any time. No identifying information was collected, and all responses remained confidential.

Measures

Physical activity was measured as self-reported total minutes completed in the past 7 days (*MinutesPA*). Anxiety was measured with the Generalized Anxiety Disorder-7 (GAD-7) questionnaire (Spitzer et al., 2006), a validated 7-item scale with total scores ranging from 0 to 21. Additional variables (sleep hours, screen time, and homework hours) were collected to consider potential confounding factors.

Results

Descriptive statistics for the two main variables appear in Table 1.

Table 1

Descriptive Statistics for Physical Activity and Anxiety

Variable	<i>M</i>	<i>SD</i>	Minimum	Q1	Median	Q3	Maximum
MinutesPA	268.08	140.61	60	147.5	255.0	392.5	500
AnxietyScore	10.58	6.24	1	7.0	11.0	14.0	21

Figure 1. Histogram of Weekly Physical Activity Minutes (*MinutesPA*).

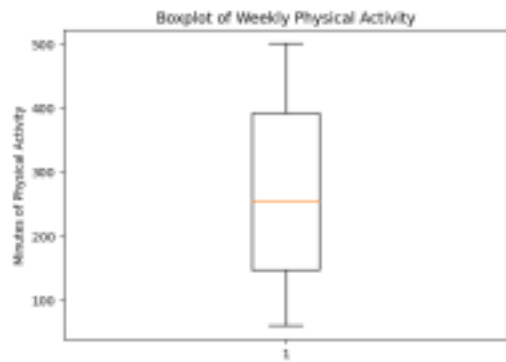
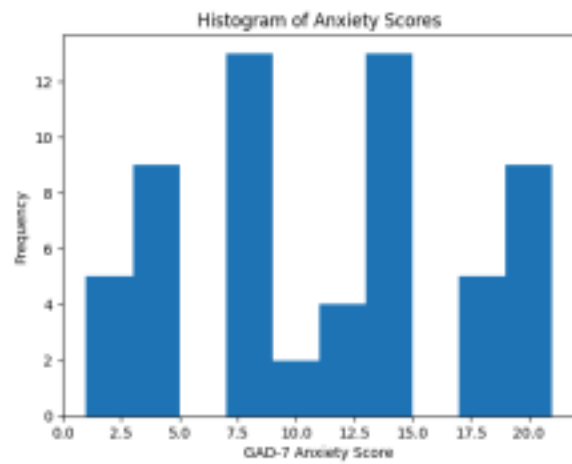


Figure 2. Boxplot of Weekly Physical Activity Minutes (*MinutesPA*).

Figure 3. Histogram of GAD-7 Anxiety Scores.



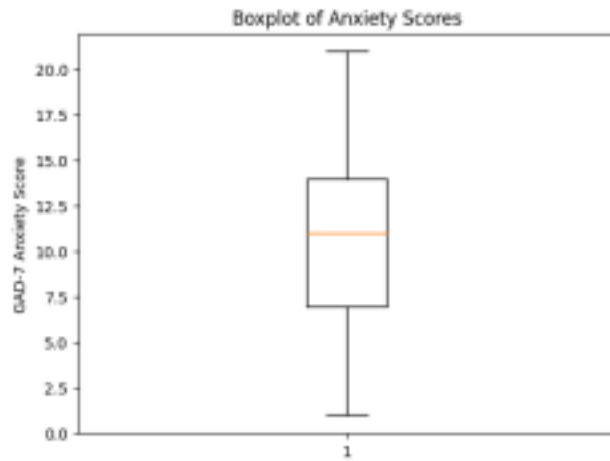
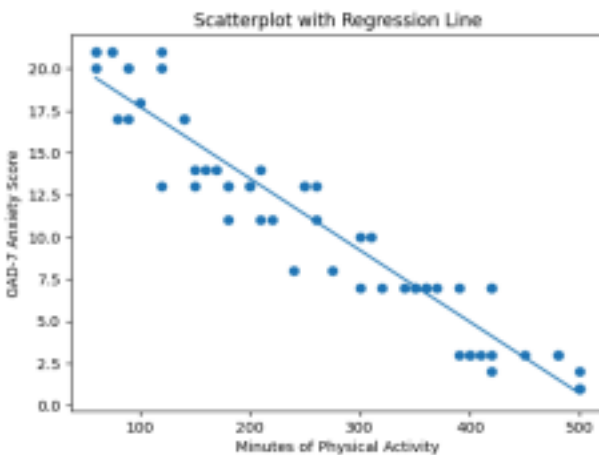


Figure 4. Boxplot of GAD-7 Anxiety Scores.

A strong negative linear relationship was evident between physical activity and anxiety (see Figure 5).

Figure 5. Scatterplot of Weekly Physical Activity and Anxiety Score with Least-Squares Regression Line.



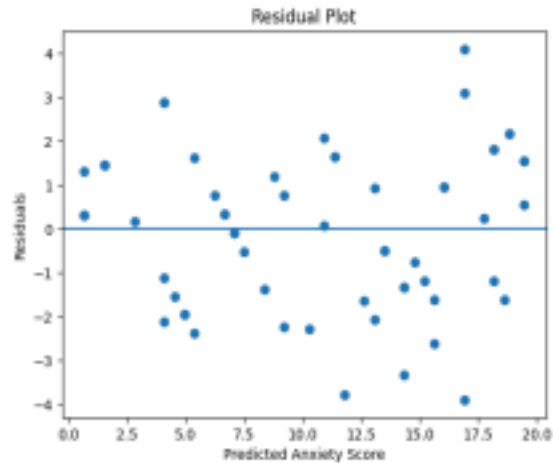
The Pearson correlation coefficient was $r = -.958$. The least-squares regression equation was:

$$\hat{y} = 21.973 - 0.04249(\text{MinutesPA})$$

$R^2 = .918$, meaning 91.8% of the variability in anxiety scores was explained by weekly physical

activity.

Figure 6. Residual Plot for the Regression Model.



Hypothesis Test for the Slope

$$H_0: \beta_1 = 0 \quad H_a: \beta_1 < 0$$

$t(59) = -25.43, p < .0001$. The null hypothesis was rejected.

The 95% confidence interval for the true slope β_1 was $[-0.04583,$

$-0.03914]$. **Additional Analysis**

A one-proportion z test showed that the proportion of students meeting CDC physical activity guidelines ($7/60 = .117$) was significantly lower than the national benchmark of .246, $z = -2.33$, $p = .010$.

Discussion

This study provides strong statistical evidence of a significant negative linear relationship

between weekly physical activity and anxiety levels among Doral Academy high school students ($r = -.958$, $t(59) = -25.43$, $p < .0001$). Students who engaged in more physical activity reported substantially lower GAD-7 anxiety scores.

These findings support the implementation of school-based initiatives to increase physical activity, such as expanded physical education, sports programs, and wellness activities, as a practical strategy to promote student mental health.

Limitations

The study relied on self-reported data, which may be subject to recall and social desirability bias. As an observational study, causation cannot be inferred. The sample was limited to one charter school, restricting generalizability.

Future Research

Future studies should include larger, multi-school samples, use objective activity trackers, and examine specific types of exercise along with additional confounding variables.

References

Centers for Disease Control and Prevention. (2024). *Physical activity guidelines for school-aged children and adolescents*. <https://www.cdc.gov/physical-activity-education/guidelines/index.htm>

Cornett, K., et al. (2024). Physical activity behaviors and negative safety and violence experiences among high school students — Youth Risk Behavior Survey, United States, 2023.

MMWR, 73. <https://doi.org/10.15585/mmwr.su7304a11>

Spitzer, R. L., Kroenke, K., Williams, J. B. W., & Löwe, B. (2006). A brief measure for assessing generalized anxiety disorder: The GAD-7. *Archives of Internal Medicine*, 166(10), 1092–1097.

<https://doi.org/10.1001/archinte.166.10.1092>

Appendix A

Survey Instrument

1. What grade are you in?
2. How many total minutes of physical activity did you complete in the past 7 days?
3. On how many days did you complete at least 60 minutes of physical activity?
4. GAD-7 Questions (7 items, scored 0–3 each).
5. Average sleep hours per night.
6. Average daily screen time (hours).
7. Average daily homework hours.

Appendix B

Representative Raw Data Sample

Grade MinutesPA AnxietyScore Sleep Screen 9 210

11 7 4 9 320 7 8 3 10 300 7 7 5 10 75 21 5 8 11 420 7 7

5 12 60 21 5 9 12 500 1 8 4

Appendix C

R Statistical Output Summary

Physical Activity and Anxiety Levels Among High School Students at Doral Academy

Teacher: Kristina Zogovic

Course: AP Statistics

Doral Academy Charter High School

Isabel Aponte, Micaela Velez, Mia Bellinski

Date: 02/25/2026

Abstract

This study investigated whether weekly physical activity predicts anxiety levels among high school students at Doral Academy in Miami, Florida. An anonymous survey asked students about total minutes of physical activity in the last seven days and used the GAD-7 anxiety questionnaire, which ranges from 0 to 21. A stratified random sample by grade level was used ($n = 60$; 15 students from each grade, 9-12). The mean weekly physical activity was 268.08 minutes ($SD = 140.61$), and the mean anxiety score was 10.58 ($SD = 6.24$). The scatterplot showed a strong negative linear relationship ($r = -0.958$). The least-squares regression equation was $\hat{y} = 21.973 - 0.04249x$. A slope t-test gave $t = -25.43$, $df = 58$, $p < 0.0001$, with a 95% confidence interval for the slope of $(-0.04583, -0.03914)$. These results provide convincing evidence that higher weekly physical activity is associated with lower anxiety scores in this sample.

I. Introduction

Schools and communities are increasingly concerned about adolescent mental health because

anxiety symptoms can affect sleep, attention, learning, and overall well-being. Physical activity is often recommended as one way to support both physical and mental health. The Centers for Disease Control and Prevention recommends that children and adolescents ages 6 to 17 get at least 60 minutes of moderate-to-vigorous physical activity each day. In this study, anxiety symptoms were measured using the Generalized Anxiety Disorder-7 questionnaire, also called the GAD-7. Since Doral Academy students experience different amounts of school stress, homework, screen time, sleep, and extracurricular activities, studying this relationship in our school population can provide useful information about student well-being.

II. Statistical Question

Among Doral Academy high school students, is there a statistically significant negative linear relationship between weekly physical activity, measured in minutes per week, and anxiety level, measured by GAD-7 score?

Hypotheses:

$H_0: \beta_1 = 0$

$H_a: \beta_1 < 0$

β_1 represents the true population slope relating weekly physical activity minutes (x) to GAD-7 anxiety score (y) for Doral Academy high school students.

III. Data Collection

The population of interest was all Doral Academy high school students in grades 9-12. We used a stratified random sample by grade level, with 15 students selected from each grade, for a total sample size of 60. Stratifying by grade helped make sure freshmen, sophomores, juniors, and seniors were all represented instead of letting one grade dominate the sample. Students

completed an anonymous survey, and no names or identifying information were collected.

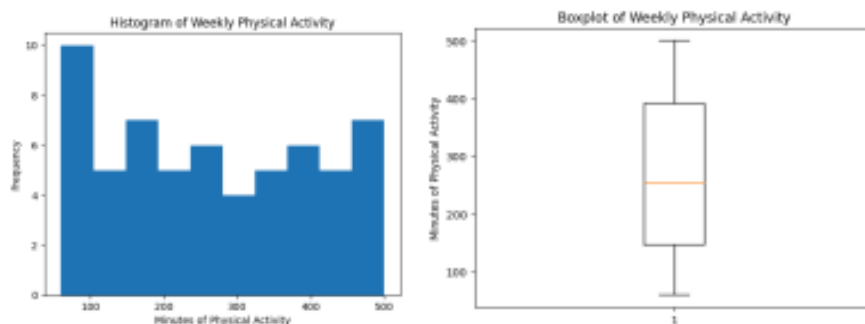
Participation was voluntary.

The explanatory variable was MinutesPA, the total number of minutes of physical activity reported in the past seven days. The response variable was AnxietyScore, the total score from questions Q4-Q10 of the GAD-7 survey. Scores could range from 0 to 21, with higher scores indicating higher anxiety. We also collected sleep, screen time, and homework time because these could be possible confounding variables. Because this was an observational survey, the results can show an association but cannot prove that physical activity causes anxiety to decrease.

IV. Data Display

Variable	Mean	SD	Five-number summary (min, Q1, med, Q3, max)
MinutesPA	268.08	140.61	(60, 147.5, 255.0, 392.5, 500)
AnxietyScore	10.58	6.24	(1, 7.0, 11.0, 14.0, 21)

Figure 1-Figure 4 shows the distributions of weekly physical activity and anxiety score. The activity data are spread from 60 to 500 minutes per week, while anxiety scores range from 1 to 21.



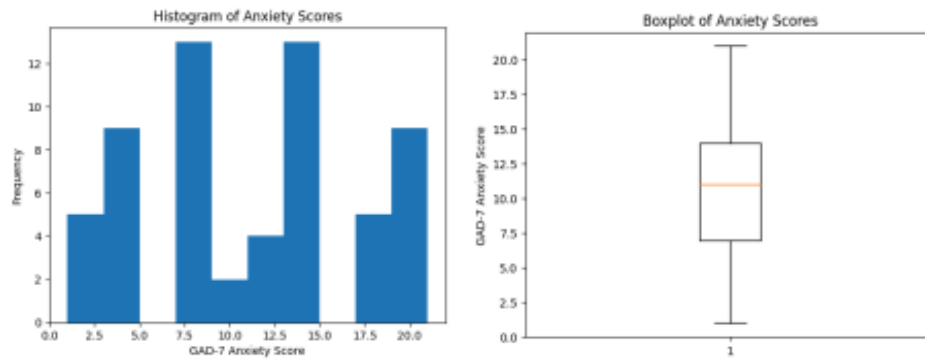


Figure 1. Histograms for weekly physical activity.

Figure 2. Boxplot for weekly physical activity.

Figure 3. Histogram for weekly GAD-7 anxiety score.

Figure 4. Boxplot for GAD-7 anxiety score.

Grade-level summary (strata check):

Grade	n	MinutesPA mean	Anxiety mean	Sleep mean	Screen mean
9	15	257.00	9.47	7.07	4.33
10	15	268.33	10.20	6.80	5.27
11	15	278.33	11.20	6.47	6.00
12	15	268.67	11.47	6.47	6.13

V. Data Analysis

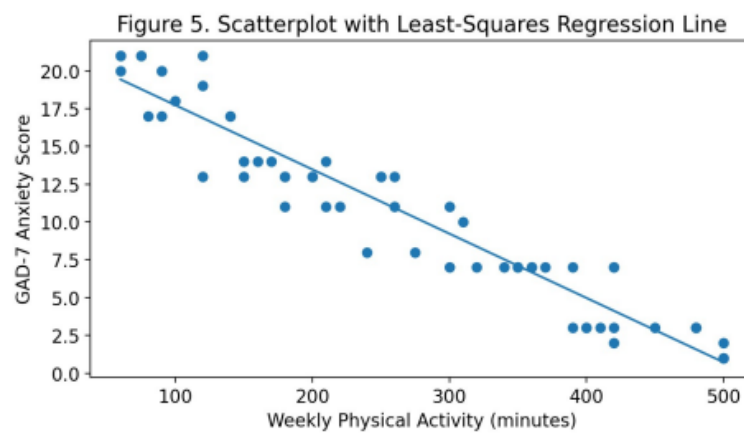


Figure 5. Scatterplot of weekly physical activity and anxiety score with the least-squares regression line.

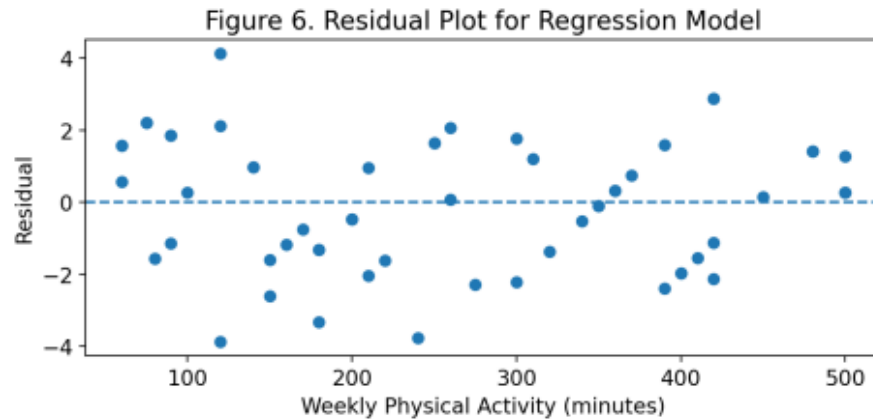


Figure 6. Residual plot for the regression model.

The correlation was $r = -0.958$, indicating a very strong negative linear association. The least squares regression model was $\hat{y} = 21.973 - 0.04249(\text{MinutesPA})$. The slope means that for each additional minute of physical activity per week, the predicted GAD-7 anxiety score decreases by about 0.04249 points. For an extra 100 minutes of activity per week, the predicted anxiety score decreases by about 4.25 points. The coefficient of determination was $R\text{-squared} = 0.918$, meaning about 91.8% of the variation in anxiety scores in this sample is explained by the linear relationship with weekly physical activity.

Conditions for inference for regression slope:

1. Linear: The scatterplot shows a clear negative linear pattern, and the residual plot does not show a strong curved pattern. This condition is met.
2. Independent: The sample was selected using stratified random sampling by grade level. Since 60 students is less than 10% of the Doral Academy high school population, the 10% condition is reasonable. This condition is met.
3. Normal: The residuals appear reasonably balanced around 0, and the sample size of 60 is

large enough for the t procedure to be robust. This condition is met.

4. Equal variance: The residual plot does not show a clear fan shape, so the variability of residuals appears fairly consistent across activity levels. This condition is met. Since the conditions were met, we used a t-test for the slope. $H_0: \beta_1 = 0$ and $H_a: \beta_1 < 0$. The test statistic was $t = -25.43$ with $df = 58$, and the p-value was less than 0.0001. Since $p < 0.05$, we reject H_0 . We have convincing evidence that higher weekly physical activity is associated with lower GAD-7 anxiety scores among Doral Academy high school students. The 95% confidence interval for β_1 was $(-0.04583, -0.03914)$, which supports the conclusion because the whole interval is below 0.

We also compared the sample to the national benchmark that 24.6% of high school students meet the seven-day aerobic guideline. In this sample, 7 out of 60 students met the guideline, so $\hat{p} = 0.1167$. For $H_0: p = 0.246$ and $H_a: p < 0.246$, the one-proportion z-test gave $z = -2.33$ and $p = 0.010$. Since $p < 0.05$, the sample proportion meeting the guideline was significantly lower than the national estimate. A 95% confidence interval for the true proportion was approximately $(0.035, 0.198)$.

VI. Conclusion

The data provide strong statistical evidence of a negative linear relationship between weekly physical activity and anxiety scores for the Doral Academy students in this sample. Students who reported more minutes of physical activity tended to have lower GAD-7 anxiety scores.

However, because the study was observational, we cannot conclude that physical activity directly caused anxiety to decrease. We can only conclude that there is a strong association in this sample.

VII. Reflection

One strength of this study was the stratified random sample by grade level, which helped make the sample more representative of the high school population. Another strength was using the GAD-7, a standardized anxiety questionnaire, instead of asking a vague question about stress. The study also included possible confounding variables such as sleep, screen time, and homework time, which helped us think more carefully about other factors that may affect anxiety.

There were also limitations. Since physical activity and anxiety were self-reported, students may have estimated incorrectly or answered in a way they thought looked better. Nonresponse bias is also possible if students who chose not to respond were different from those who participated. The relationship was unusually strong for behavioral data, so future studies should use a larger sample and possibly collect data across several schools. A stronger follow-up study could compare students over time or use an experimental design where students follow different activity plans, which would make it easier to study cause and effect.

VIII. Citations

American Statistical Association. (2025). ASA Statistics Project Competition for Grades 7-12.

<https://www.amstat.org/education/asa-statistics-project-competition-for-grades-7-12>

Centers for Disease Control and Prevention. (2024). Physical activity guidelines for school-aged children and adolescents. <https://www.cdc.gov/physical-activity-education/guidelines/index.htm>

Cornett, K., et al. (2024). Physical activity behaviors and negative safety and violence experiences among high school students - Youth Risk Behavior Survey, United States, 2023.

Morbidity and Mortality Weekly Report. [Physical Activity Behaviors and Negative Safety and](#)

[Violence Experiences Among High School Students — Youth Risk Behavior Survey, United States, 2023 | MMWR](#)

Spitzer, R. L., Kroenke, K., Williams, J. B. W., & Lowe, B. (2006). A brief measure for assessing generalized anxiety disorder: The GAD-7. Archives of Internal Medicine.

<https://pubmed.ncbi.nlm.nih.gov/16717171/>

Appendix A: Survey Instrument

Students answered: grade level, total minutes of physical activity in the last seven days, number of days with at least 60 minutes of activity, seven GAD-7 anxiety questions scored from 0 to 3, average sleep hours, average screen time hours, and average homework hours.

Appendix B: Hand Calculations

$r = -0.958$; $y\text{-hat} = 21.973 - 0.04249x$; $t = -25.43$; $df = 58$; $p < 0.0001$; 95% CI for $\beta_1 = (-0.04583, -0.03914)$. For the proportion test, $p\text{-hat} = 0.1167$, $z = -2.33$, and $p = 0.010$.

Appendix C: R Code and Raw Data

R code and the full raw data file are included in the project repository and RPub page:

<http://rpubs.com/isabelaponte/1401974>